Chapter 4

A “Mostly Normal” Introduction to Hypothesis Testing

Introduction

A **hypothesis test** uses an estimator to make a decision about a population parameter.

This chapter will focus on the principles and mechanics of deriving a hypothesis test.

4.1

Getting Started, Some Intuition

A story

Suppose $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, and we observe $\bar{x} = 3.7$. We wish to decide whether $\mu \leq 3$ or $\mu > 3$ based on our observed data.

We will set up two hypotheses, H_0 and H_1, that are written as

$$H_0 : \mu \leq 3 \quad \text{versus} \quad H_1 : \mu > 3.$$

Definition 4.1.1

H_0 is called a **null hypothesis**.

- It is the hypothesis that is assumed to be true, initially.

H_1 is called an **alternative hypothesis**.

- This hypothesis states what we will conclude to be true if we are swayed by strong evidence from the sample.

More thoughts

Even if $\bar{x} > 3$, we may not want to conclude H_1 is true.

We are assuming that H_0 is true. How much evidence is needed to convince us that H_0 is incorrect and H_1 is true?

4.2

Hypotheses: Simple or Composite?

Simple Hypotheses

A **simple hypothesis** completely specifies the data's distribution.

- E.g., $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, 1)$ and the null hypothesis is $H_0 : \mu = 3$.
- If H_0 is true, then the random sample comes from a $N(3, 1)$ distribution.

Composite Hypotheses

A **composite hypothesis** is one that does not completely specify the data's distribution.

- E.g., $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, 1)$ and the null hypothesis is $H_0 : \mu \leq 3$. If H_0 is true, then the random sample could come from a $N(-1, 1)$ distribution or a $N(-79, 1)$ distribution.
- E.g., $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and the null hypothesis is $H_0 : \mu = 3$. If H_0 is true, then the random sample could come from a $N(3, 1)$ distribution or a $N(3, 27)$ distribution.

4.3

Errors in Hypothesis Testing for a Simple Null Hypothesis

Setting the stage

H_0 is either true or false.

We assume H_0 is true when we start the testing process, but will conclude that H_1 is true if the data strongly support it enough.

There are many types of errors we can make when performing hypothesis testing.

Testing error types table

4.3.1

Level of Significance: α

Definition 4.3.1

For a *simple* null hypothesis, the **level** of a test, α, is the probability of making a Type I error.

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \text{ when it's true}).$$

Example

Let $X_1, X_2, \ldots, X_{10} \overset{i.i.d.}{\sim} N(\mu, 1)$.

- Test $H_0 : \mu = 3$ versus $H_1 : \mu > 3$.
- Reject H_0 when $\bar{X} > 4$ (for illustration purposes).

What is the level of the test?

Example (cont)

Example (cont)

4.4

Finding a Test: A Simple Null Hypothesis

A simple testing example

Let $X_1, X_2, \ldots, X_{10} \overset{i.i.d.}{\sim} N(\mu, 1)$.

Construct a test of $H_0 : \mu = 3$ versus $H_1 : \mu > 3$ based on the sample mean, $\bar{X}$, such that the level of the test is $\alpha = 0.05$.

Some thoughts:

- If H_1 is true, then large $\bar{X}$ (more than 3) support H_1.
- The *form* of our rejection rule is, "Reject H_0, in favor of H_1, if $\bar{X} > c$".

Simple example (cont)

Simple example (cont)

Some definitions

The **rejection rule** is the rule we used to decide whether we reject H_0.

The **rejection region** (or set) is the set of values that will cause us to reject H_0.

The rejection rule and rejection region will generally change depending on the level of the test.

Note

The form of the rejection rule is determined by the alternative hypothesis, but the direction of the alternative hypothesis may not translate to the same direction in the decision rule.

E.g., let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$.

We test $H_0 : \lambda = 3$ versus $H_1 : \lambda > 3$ using $\bar{X}$.

What should our rejection rule look like?

Note (cont)

$\bar{X}$ is an estimator of the mean of the distribution.

The mean of the distribution is $1/\lambda$

Thus, a larger value of λ is linked to smaller values of $\bar{X}$.

The form of the rejection rule for this setting is, "Reject H_0, in favor of H_1, if $\bar{X} < c$, where c is chosen to give a level α test.

4.5

Finding a Test: A Composite Null Hypothesis

Motivating example

Let $X_1, X_2, \ldots, X_{10} \overset{i.i.d.}{\sim} N(\mu, 1)$.

Construct a test of $H_0 : \mu \leq 3$ versus $H_1 : \mu > 3$ based on the sample mean, $\bar{X}$, such that the level of the test is $\alpha = 0.05$.

Some intuition

The form of the rejection rule is the same.

H_1 being true means that the sample mean should be large.

We will reject H_0, in favor of H_1, when $\bar{X} > c$, where c is chosen to control the Type I error.

Deriving the rejection rule

Notation

We denote the parameter space of a generic parameter θ by Θ.

θ must be an element of Θ, i.e., $\theta \in \Theta$.

Let Θ_0 be some subset of Θ.

We want to test

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta \setminus \Theta_0.$$

More details

Recall that the backslash here is “setminus” notation.

$\Theta \setminus \Theta_0$ is everything left in Θ after removing all the elements of Θ_0.

Alternatively, this set could be written as

$$\Theta \setminus \Theta_0 = \Theta \cap \Theta_0^c.$$

- Let $\Theta = \mathbb{R}^+ = (0, \infty)$ and $\Theta_0 = (1, \infty)$.
- Determine Θ_0^c and $\Theta \setminus \Theta_0$.

Definition 4.5.1 (Size of a test)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution that depends on a parameter $\theta \in \Theta$.

Let $\Theta_0 \subset \Theta$.

The **size** of the test of

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \Theta \setminus \Theta_0$$

is denoted by α, where α is the maximum probability of making a type I error under the null hypothesis, i.e.,

$$\alpha = \max_{\theta \in \Theta_0} P(\text{Reject } H_0 \text{ when the parameter is } \theta).$$

- Technically, we should use $\sup$ in case the maximum only occurs as a limit.

Many authors (including ours) don't make this distinction, but the well-known *Statistical Inference, 2nd edition* text by Casella and Berger does.

Size versus level of a test

A test is of level α if

$$\max_{\theta \in \Theta_0} P(\text{Reject } H_0 \text{ when the parameter is } \theta) \leq \alpha.$$

A test is of size α if

$$\max_{\theta \in \Theta_0} P(\text{Reject } H_0 \text{ when the parameter is } \theta) = \alpha.$$

The set of level α tests contains the set of size α tests.

Size discussion

The maximum probability of a type I error is taken over all values of the parameter possibilities possible under H_0.

If $H_0 : \mu \leq 3$, like our previous example, then

$\alpha =$

Size discussion (cont)

If we say $P(\text{Reject } H_0; \mu)$, i.e., the probability that we reject H_0 when the parameter is μ, does not indicate that any error has been made.

However, if we say $\max_{\theta \in \Theta_0} P(\text{Reject } H_0; \mu)$, then we are considering the probability of a type I error since we are assuming parameter values of θ only for values of θ consistent with H_0.

By setting α using the maximum, we are controlling the highest probability of making a type I error.

Example 4.5.1

Let $X_1, X_2, \ldots, X_{10} \overset{i.i.d.}{\sim} N(\mu, 1)$.

We want to find a test of size $\alpha = 0.05$ when testing

$$H_0 : \mu \leq 3 \quad \text{versus} \quad H_1 : \mu > 3$$

based on the sample mean.

Example 4.5.1 (cont)

Example 4.5.1 (cont)

Example 4.5.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

We will test the hyptheses

$$H_0 : \mu \geq \mu_0 \quad \text{versus} \quad H_1 : \mu < \mu_0,$$

for a fixed value of $\mu_0 \in \mathbb{R}$.

Find a test of size α.

Example 4.5.2 (cont)

Example 4.5.2 (cont)

Example 4.5.2 (cont)

Example 4.5.2 (cont)

4.5.1

Where There’s an α, There’s a β.

Definition 4.5.2

For a simple alternative hypothesis, β denotes the probability of a type II error, i.e.,

$$\beta = P(\text{Type II error}) = P(\text{Fail to reject } H_0 \text{ when it's false}).$$

Definition 4.5.3

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution that depends on a parameter θ that lives in a parameter space Θ.

Let $\Theta_0 \subset \Theta$.

Consider testing

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta \setminus \Theta_0.$$

β is the maximum probability of making a Type II error, i.e.,

$$\beta = \max_{\theta \in \Theta \setminus \Theta_0} P(\text{Fail to reject } H_0 \text{ when the parameter is } \theta).$$

An equivalent definition

Definition 4.5.4

The **power** of a test $1-\beta$, where

$$1-\beta = \min_{\theta \in H_1} P(\text{Reject } H_0; \theta).$$

Making a connection

The rejection region of a test is defined by:

- The test statistic used.
- The form of the alternative hypothesis.
- The size or level of the test.

Once these three testing characteristics are chosen, the probability of a Type II error is “locked in” to a specific number.

Example 4.5.3

Let $X_1, X_2, \ldots, X_{10} \overset{i.i.d.}{\sim} N(\mu, 1)$.

We wish to test

$$H_0 : \mu = 3 \quad \text{versus} \quad H_1 : \mu > 3,$$

based on the sample mean, $\bar{X}$.

When $\alpha = 0.05$, we previously determined that the resulting rejection rule is "Reject H_0 if $\bar{X} > 3.52$".

Determine β.

Example 4.5.3 (cont)

Example 4.5.3 (cont)

Additional points of connection

- We cannot make a Type I and Type II error at the same time.
- α and β are linked together.
  - Fixing a small α will increase β.
- Is there a way to control both α and β?

4.6

Non-Normal Distribution Hypothesis Tests

Example 4.6.1

Suppose that $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$.

Derive a test of size α for the hypotheses

$$H_0 : \lambda \leq \lambda_0 \quad \text{versus} \quad H_1 : \lambda > \lambda_0,$$

for some fixed $\lambda_0 > 0$.

Base your test on $X_{(1)}$, the minimum value of the sample.

Example 4.6.1 (cont)

Example 4.6.1 (cont)

Example 4.6.1 (cont)

Example 4.6.1 (cont)

Example 4.6.2

Suppose that $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$.

Derive a test of size α for the hypotheses

$$H_0 : \lambda \leq \lambda_0 \quad \text{versus} \quad H_1 : \lambda > \lambda_0,$$

for some fixed $\lambda_0 > 0$.

Base your test on $\bar{X}$, the sample mean. Leave your test in terms of a χ^2-critical value.

Example 4.6.2 (cont)

Example 4.6.2 (cont)

Example 4.6.2 (cont)

Example 4.6.2 (cont)

4.7

Power Functions

Power functions

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} F_\theta$.

We wish to test

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta \setminus \Theta_0.$$

A **power function** is useful for comparing two different size α tests.

Definition 4.7.1

The **power function** for a hypothesis test concerning θ is

$$\gamma(\theta) := P(\text{Reject } H_0; \theta),$$

for all $\theta \in \Theta$.

Power vs power function

The power of a test and the power function are not the same thing, but they are related.

Power vs power function (cont)

Power vs power function (cont)

Example 4.7.1

In Example 4.5.2, we assumed $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and derived a test of $H_0 : \mu \geq \mu_0$ versus $H_1 : \mu < \mu_0$.

Our rejection rule was "Reject H_0 if $\bar{X} < \mu_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}}$."

Determine the power function of the test.

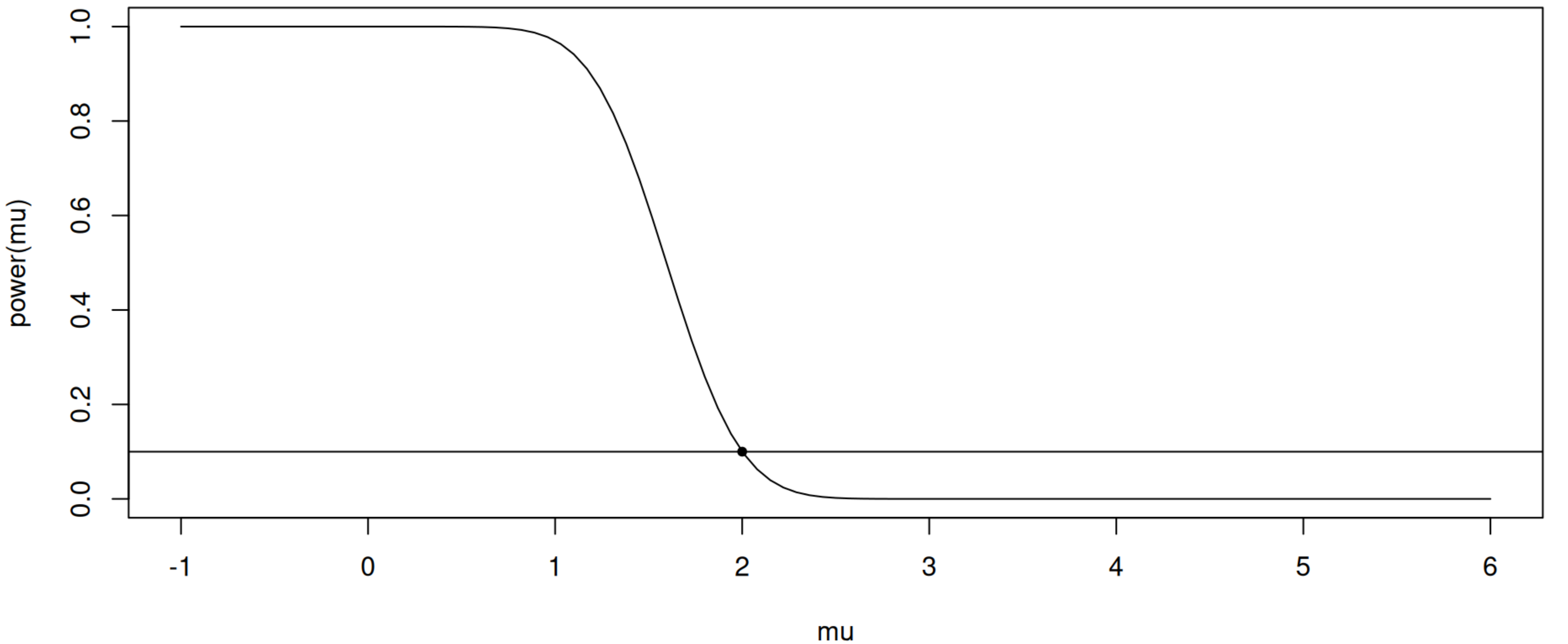
Example 4.7.1 (cont)

Example 4.7.1 (cont)

Example 4.7.1 (cont)

Example 4.7.1 (cont)

Example 4.7.1 (cont)



Example 4.7.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\lambda)$.

In Section 4.6, we developed two different hypothesis tests of size α for

$$H_0 : \lambda \leq \lambda_0 \quad \text{versus} \quad H_1 : \lambda > \lambda_0.$$

The decision rule of Test 1 is "Reject H_0 if $X_{(1)} < -\frac{1}{n\lambda_0}\ln(1-\alpha)$".

The decision rule of Test 2 is "Reject H_0 if $\bar{X} < \frac{\chi^2_{1-\alpha,2n}}{2n\lambda_0}$".

Example 4.7.2 (cont)

Example 4.7.2 (cont)

Example 4.7.2 (cont)

Example 4.7.2 (cont)

Example 4.7.2 (cont)